Genetic Programming Classifier Report

COS314 Project

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Abstract

This report summarizes the implementation and comparison of two Genetic Programming classifiers on the Breast Cancer Wisconsin dataset. One variant evolves arithmetic expressions, while the other evolves tree-based decision

1 Introduction

Genetic Programming (GP) evolves classifiers by applying selection, crossover, and mutation to a population of program trees. This project implements two GP variants in Java and evaluates them on a standard breast cancer dataset.

1.1 Experimental Setup

Each algorithm was executed for 30 independent runs using seeds given. The best-performing run was selected based on training fitness and saved. Saved in results directory. The seed used for the demonstrated Decision Tree GP run was `0`, also the case for the demonstrated Arithmetic GP.

2 Methods

The two variants are:

- arithmetic-expression GP, which evolves symbolic numerical formulas
- decision-tree GP, which evolves logical tree-based classifiers

Both variants use the same dataset and evaluation metrics. Each variant is ran 30 times with the same inputted seeds. The program saves one result file per run under `results/`.

2.1 GP Parameters

Parameter	Value
Population Size	200
Maximum Generations	100
Crossover Rate	0.8
Mutation Rate	0.2
Maximum Depth	5
Tournament Size	5
Selection Method	Tournament Selection
Initialisation Method	Ramped Half-and-Half
Fitness Function	Training Accuracy

3 Results

The main comparison is based on accuracy, F-measure, and runtime. In the tested runs, the decision-tree GP achieved higher test F-measure, while the arithmetic GP reached a bit higher training accuracy.

Algorithm	Training (%)	Test (%)	F-measure	Runtime (s)
Decision Tree GP	95.08	53.49	0.3750	0.49
Arithmetic GP	95.08	50.00	0.3944	0.35

A two sample t-test on test accuracies found no statistically significant difference at the 0.05 level for this experimental setup.

3.1 Program Output

During training, the program prints the best individual of each generation together with its training accuracy.

4 Discussion

The decision-tree variant appears to generalize better on the test split, while the arithmetic variant fits the training set more closely. Further work should include cross-validation, parameter tuning and comparison with standard machine learning strategies.

5 Although

limitation observed during testing was occasional overfitting, where individuals achieved high training accuracy but weaker test performance. This was more noticeable in the arithmetic GP variant.

6 Conclusion

This project implements two GP classifiers and compares their performance on breast cancer classification.